



BEST PRACTICE SERIES

Planning and Scheduling (Strategic)

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

Planning and Scheduling (PS) is one of the core functions of the maintenance and repair system. It plays a critical role in the development, implementation and execution of a repair-before-failure philosophy and supports the goal of achieving a high percentage of planned and scheduled repairs. Planning & Scheduling must define the maintenance strategy, transform it into clear and effective plans and activities, schedule them through to execution, and control the results by developing the most appropriate corrective management action plans.

Planning and Scheduling manages all of the various maintenance-related downtime scenarios throughout a machine's lifecycle: Condition Monitoring, Unscheduled and Scheduled repairs, Preventive Maintenance and Planned Component Replacements. See Figure 1.

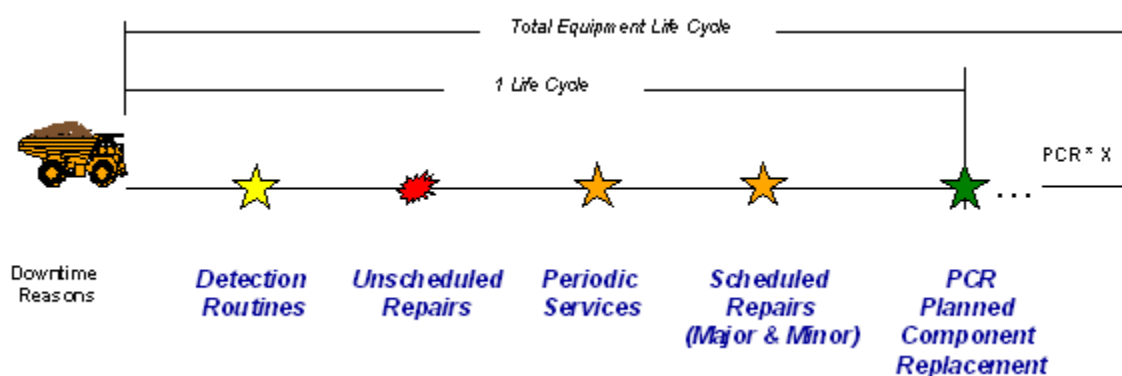


Fig. 1

These maintenance and repair activities must be planned and executed in coordination with the labor, machine, parts, and resource availability. Effective Planning and Scheduling ultimately achieves a balance between M&R required activities, resource availability and capability, and overall fleet availability. It's a matter of completing the required monthly M&R activities within the target fleet unavailability time reserved for maintenance and repair.

2.0 Best Practice Description

Description Outline

2.1 Planning & Scheduling Overview

2.2 Planning & Scheduling Process

- 2.2.1 Strategy
- 2.2.2 Records & Recordkeeping
- 2.2.3 Planning
- 2.2.4 Scheduling
- 2.2.5 Technical Support
- 2.2.6 Availability Management
- 2.2.7 Performance Evaluation

2.1 Planning & Scheduling Overview

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Planning and Scheduling functions can be categorized as tactical...**“what”** should we do?”, technical...**“how”** should we do it?” and logistical...**“when”** should we do it?”. Planning must first provide the “What” by transforming the Maintenance Strategy into actionable plans supported by clear goals and objectives. Scheduling follows with the “when” by defining the actual schedules of the plan activities, when to execute them and triggering the allocation of needed resources. The what, how and when can be represented three major areas or columns, the Tactical, the Technical and the Logistical. See Figure 2.

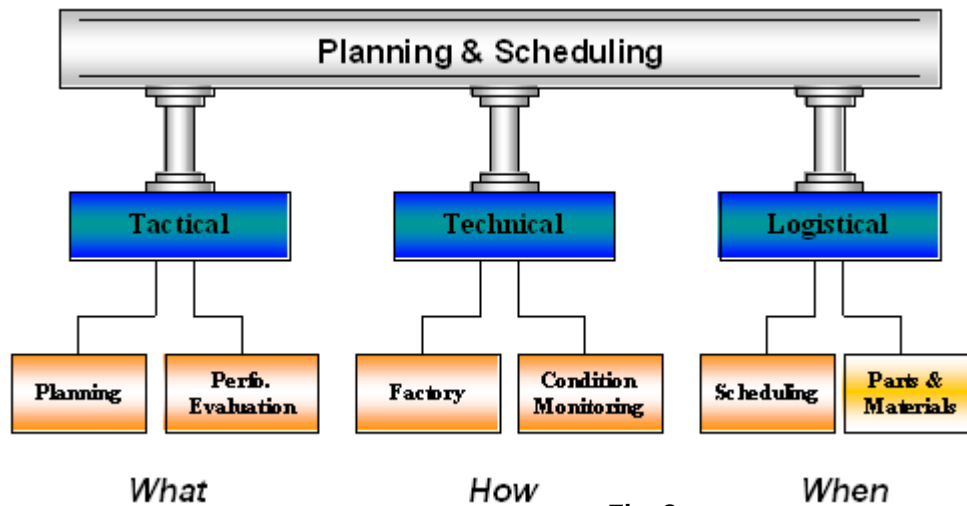


Fig. 2

The planning organization must first define their overall goals and objectives before a maintenance strategy or plan can be developed. We offer four main goals and objectives:

- Provide tactical & logistical support to the entire organization.
- Provide updated Short, Medium & Long Term maintenance plans.
- Track and support the mine production plans to balance the maintenance activities within the availability goals (manage fleet availability).
- Guide the M&R organization with updated performance/execution information, KPI's and management recommendations.

There are many inputs that must be considered when developing the maintenance strategy and plan in support of the goals and objectives. The strategy, which yields the guidelines for each maintenance activity, will be unique depending upon site-specific details. For example, the production plan, fleet characteristics, dealer maintenance contractual definitions, and application severity will dictate available fleet downtime, dealer maintenance responsibilities, maintenance activities required, etc... this is the **“what”** portion of the strategy. Other inputs such as maintenance facilities, budget, parts, and labor availability will define the logistical **“how”** and **“when”** portions of the maintenance strategy. Strategy development will be covered in further detail in section 2.2.1.

The Planning and Scheduling organization is ultimately responsible for the support, execution, management, and success of the overall maintenance program while still accommodating the

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production plan. The responsibility is quite extensive. Therefore, in support of the maintenance strategy and plan, the implementation of multiple other process and organizations will be required. These processes will provide both the inputs and execute the outputs of Planning and Scheduling. For example, the Condition Monitoring (CM) groups' findings will generate the input that generates/adjusts the short, medium, and long-term maintenance plans. At the same time, P&S will define (output) the activities to be executed by the CM group. In other words, the P&S group defines the CM activities that feed directly back into the P&S information input - creating a closed process loop. The same relationship exists for the Preventive Maintenance, Repair Management, Backlog Management, and Parts groups. The following graphic illustrates that Planning and Scheduling is truly the brain of the operation sitting at the center of the supporting disciplines, both receiving information and defining guidelines and activities for each. See Figure 3.

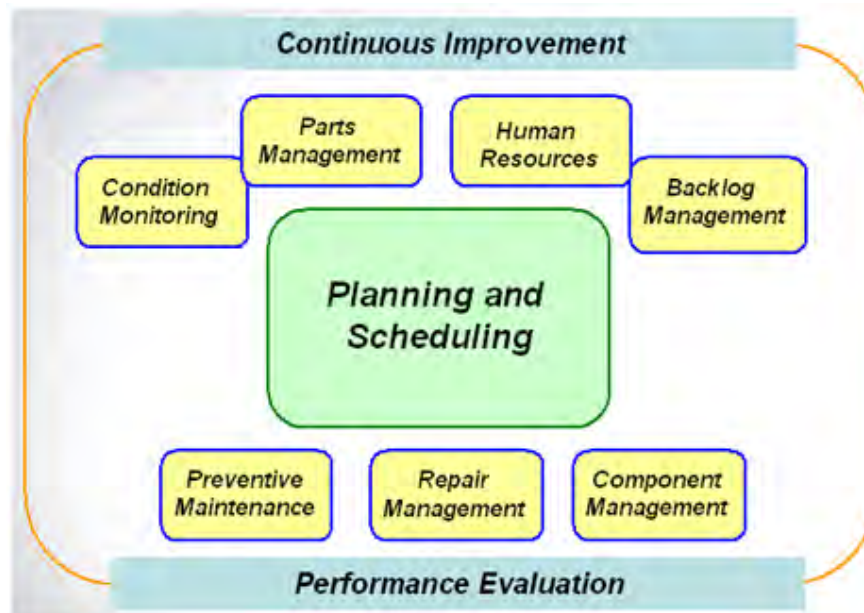


Fig. 3

Planning's main function is to translate the maintenance strategy into continuously updated maintenance plans.

The plan then drives the functions of:

1. Event scheduling
2. Availability management
3. Labor/resource management
4. Record keeping
5. Provide technical support
6. Performance evaluation
7. Identify and lead continuous improvement plans

2.2 Planning & Scheduling Process

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2.2.1 Strategy

The basis for any successful Planning and Scheduling process is a documented and complete strategy. As mentioned, the strategy will be the governing reference document that defines **what** the organization intends to accomplish and a plan of **how** the strategy will be executed. Robust strategies include both the high level P&S objective and description but also outlines of the process approach, sequencing, duties, and functions. The following is an example of a well-defined P&S strategy index (See Table 1). Notice that the strategy includes the description for the specific P&S processes and also the supporting functions mentioned above (condition monitoring, parts management, etc):

INDEX	
PREFACE	
REVISIONS AND ALTERATIONS CONTROL	
1. OBJECT	
2. INTRODUCTION	
3. LOGICAL SEQUENCE OF PLANNING AND SCHEDULING	
3.1 PREPARATION FOR IMPLEMENTATION OF PM AND OF COMPONENTS PROGRAM	
3.2 SCHEDULING OF SERVICES TO BE EXECUTED IN THE SERVICE SHOP	
3.3 UNEXPECTED FAILURES AND NON SCHEDULED REPAIRS	
3.4 DATA ENTER IN THE MAINTENANCE CONTROL SYSTEM THE W/O CLOSING INFORMATION	
4. DETAILS OF THE PROCESS	
4.1 PLANNING ACTIVITIES	
4.2 PREPARATION OF MAINTENANCE STRATEGICS	
4.3 PLANNING AND SCHEDULING OF EQUIPMENT STOP FOR MAINTENANCE	
4.4 CONDITION MONITORING	
4.5 PARTS MANAGEMENT	
4.6 COMPONENT MANAGEMENT	
4.7 TECHNICAL SUPPORT AND PROBLEM MANAGEMENT	
4.8 NECESSARY RESOURCES FOR MAINTENANCE EXECUTION	
4.9 NECESSARY RESOURCES FOR INFORMATION PROCESSING	
4.10 INFORMATION MANAGEMENT	
4.11 RESULTS AND EVALUATION OF FLEET PERFORMANCE	
4.12 RESULTS AND EVALUATION OF CONTRACT'S PERFORMANCE	
5. TRAINING	
5.1 RESPONSIBILITIES	
5.2 IDENTIFICATION OF NEEDS	
5.3 PROGRAMMING OF COURSES	
5.4 EVALUATION OF RESULTS	
6. JOB DESCRIPTIONS	
6.1 PLANNING AND SCHEDULING SUPERVISOR	
6.2 SCHEDULER	
6.3 MAINTENANCE CLERK	
6.4 FLEET ANALYST	
6.5 CONTRACT PARTS ASSISTANT	

Table 1

One tool available to help develop the P&S strategy is the "Maintenance Strategy Reference (MSR)." MSR is a tool available to specifically define a model-specific maintenance strategy. Dealer personnel use the tool to document the site profile, equipment profile, risks, etc. in order to develop the condition monitoring, preventive maintenance, and roles & responsibilities strategy to best maintain that machine. However, this is only a portion of the overall P&S organization strategy.

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Now that the P&S strategy is defined, the organization can begin developing the overall P&S plan for each maintenance sub-function (PM, Condition Monitoring, Backlogs, etc). These functions will be carried out by the various maintenance areas either owned or in partnership with the P&S group. The plan must define three elements for each maintenance function:

1. **Functions and routines**
2. **Resources required**
3. **Expected results**

The overall process diagram below shows the seven areas that must work together and contribute to the execution and evaluation of each maintenance function (Preventive Maintenance, Condition Monitoring, Backlogs, etc).

1. **Planning:** planning of all the maintenance activities
2. **Data Collection:** Implement the data collection systems and collection of the Production and M&R data (i.e. machine availability, delays, stoppages, PM intervals, backlogs pending, etc.)
3. **Scheduling:** management and triggering of machines and resources to execute the plan.
4. **Logistics:** management of the parts and tools required to execute the plan.
5. **Human Resources:** management of the labor required to execute the plan.
6. **Technical Support:** On or off site technical resources required to resolve and communicate maintenance specifications.
7. **Control:** Utilization of the data collection results to evaluate the M&R organization effectiveness.

Immediately following the General Process map below (Figure 4), an example process review will be shown for the Preventive Maintenance function. This specific example will demonstrate how the three elements (routines, resources, results) are defined for PM's and how the seven areas are required in support of the PM activity.

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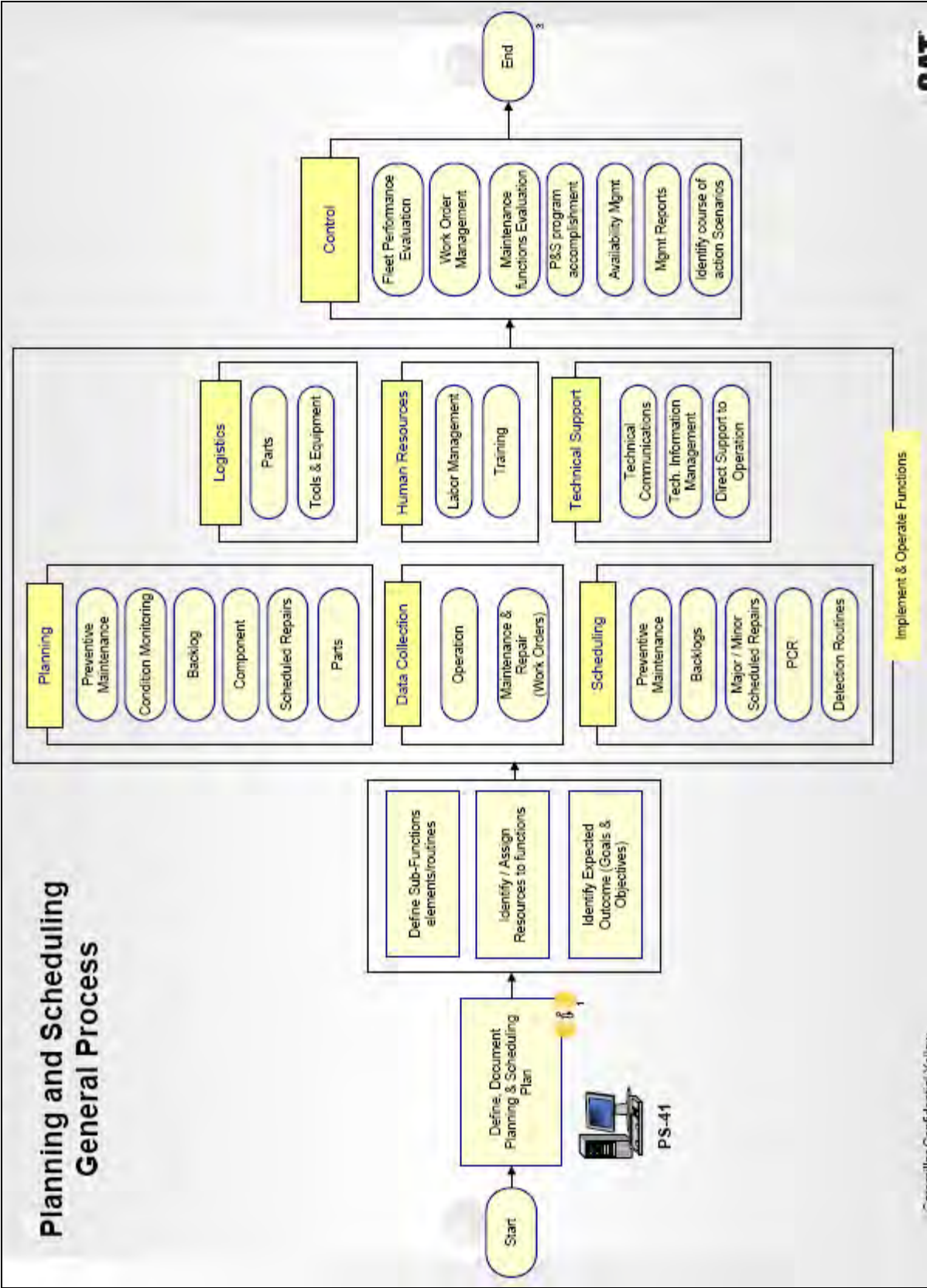
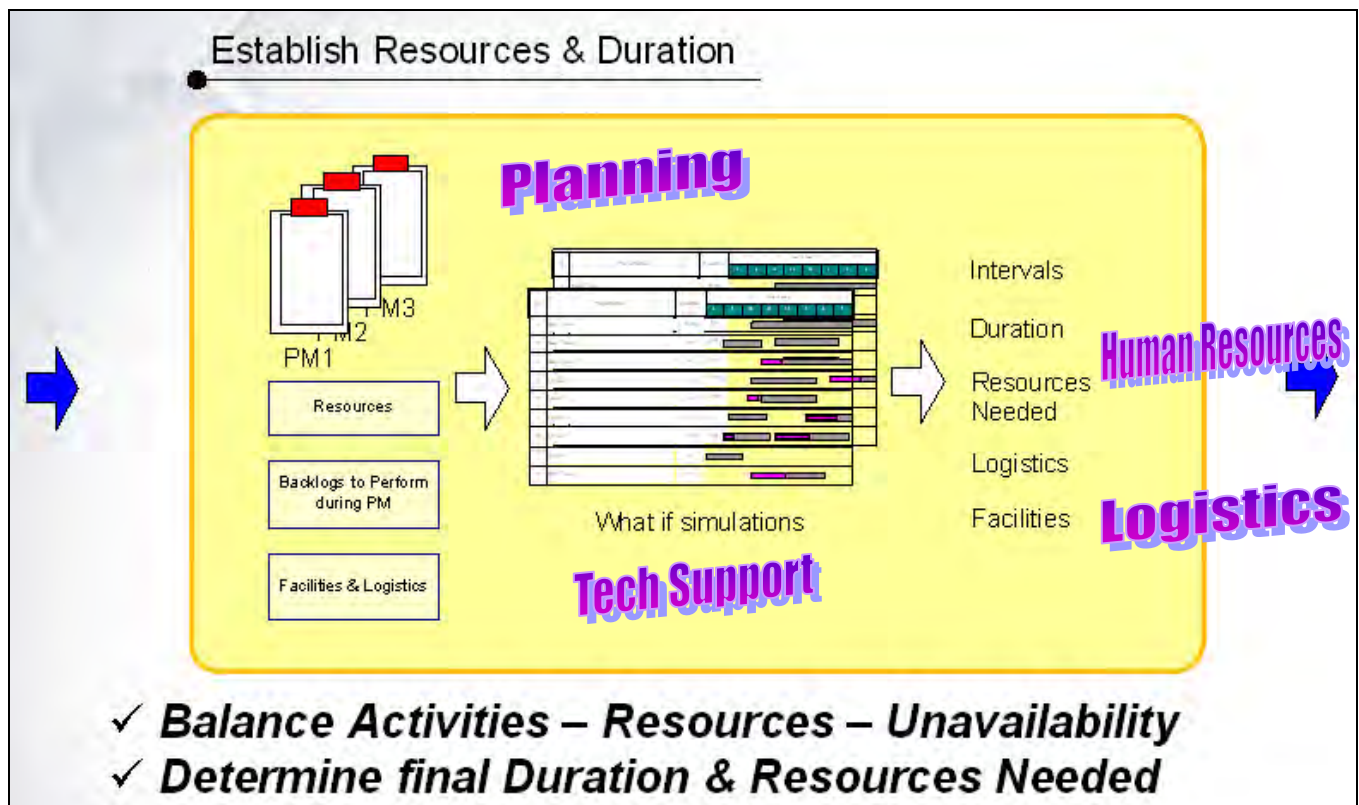
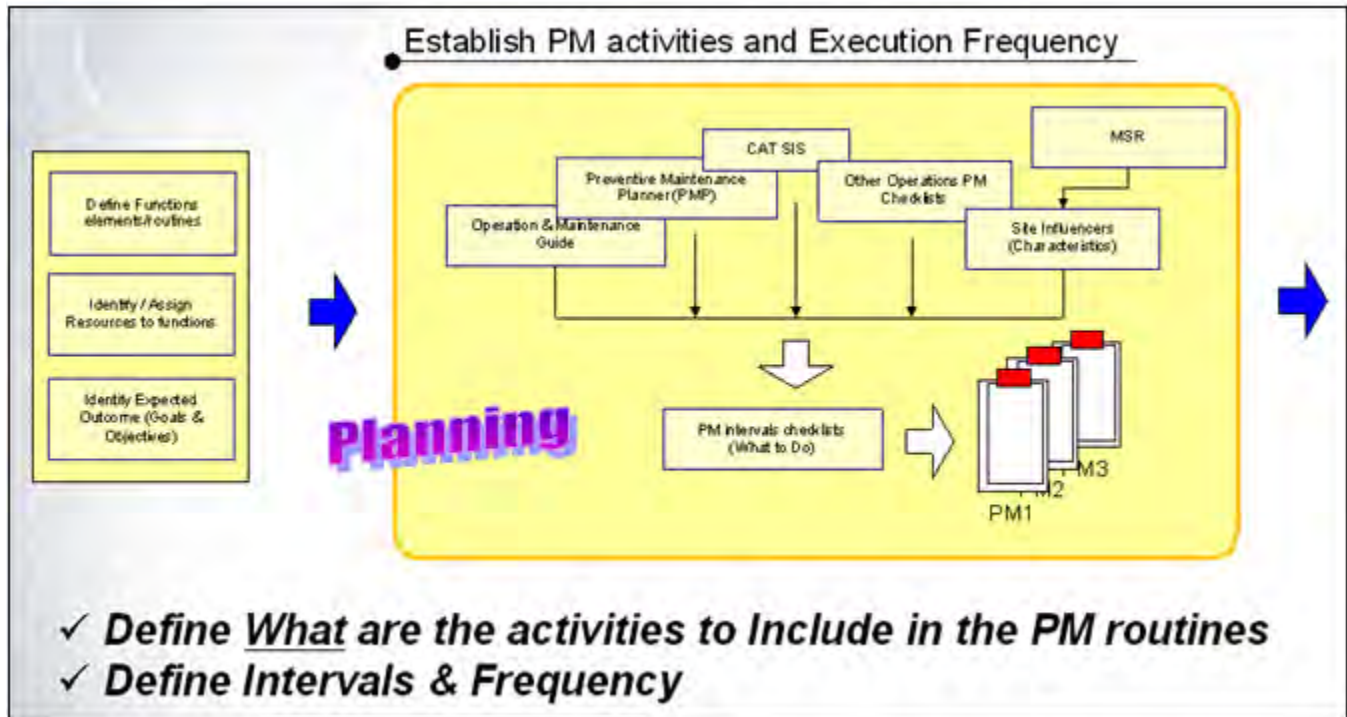


Fig. 4

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Example of the Plan definition for Preventive Maintenance (one of the multiple maintenance functions/activities that must be defined by the plan) and elements of the seven areas required for execution of the PM activity:



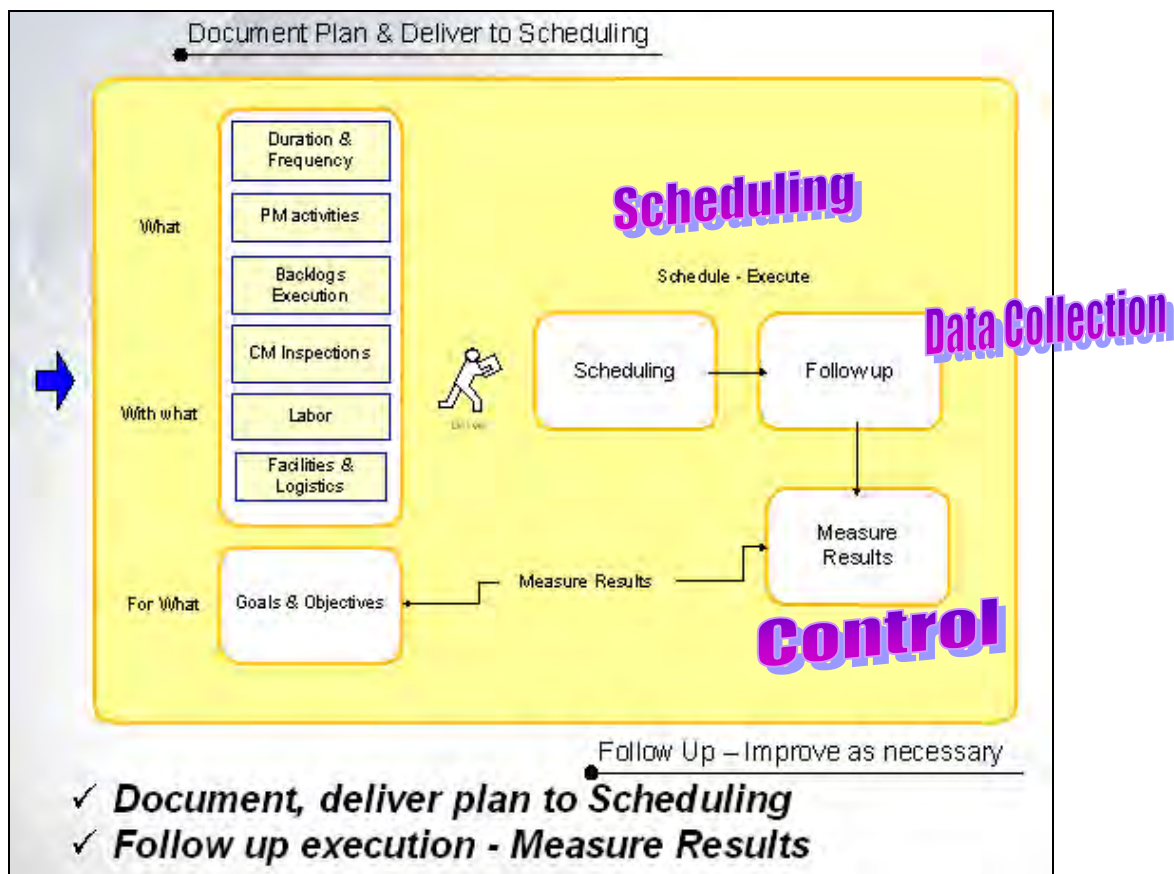
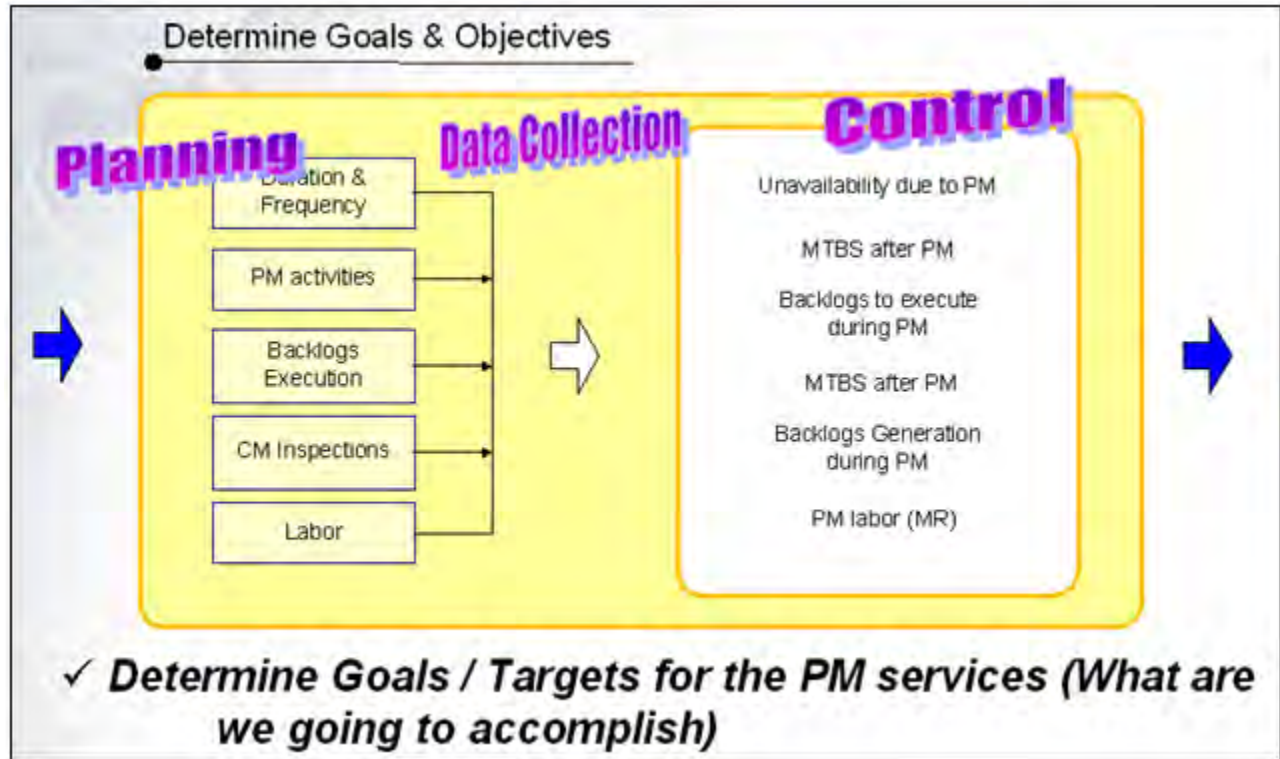
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2.2.2 Records & Recordkeeping

Timely, accurate, and complete data is critical for effective Planning and Scheduling. The planning function starts with machine performance and work status data from both the production and maintenance organizations. Reports and data from these groups drive the planning decision process:

Sources of data/reports:

1. Mine Dispatch – Machine availability and downtime data.
2. Field Service – Shift reports and work completion status.
3. Service Shop – Shift reports and completed work orders.
4. Tire shop – tire availability, shift progress, and completed orders.

Remember that one of the major P&S goals is to track and support the mine production plans by balancing the maintenance activities within the unavailability goals (manage fleet availability). Planning must begin with accurate machine performance data showing machine downages both maintenance and non-maintenance related. Only with this data can Planning evaluate the ongoing M&R completed work versus fleet availability and adjust the maintenance plans accordingly. There are two sources for this data; the dispatch log from operations and the work orders/shift log from maintenance. Since the two systems are independent, frequent meetings must occur between operations and maintenance to reconcile these two databases. The operations and maintenance organizations must agree on when the machine was down and for what reasons...mainly maintenance or non-maintenance related which will determine contractual availability. For best results this reconciliation meeting should occur daily while the events are current and easily recalled.

Now that the P&S group knows available time for maintenance activities, it must have the maintenance data showing work completion rate in accordance with the plan and schedule. P&S must have a system in place to compare shift log reports with work order entry to determine when and what type of work is being completed. P&S uses this information to determine time spent for each maintenance activity (PM's, detection routines, scheduled and unscheduled repairs, etc) and determines if the plan and schedule must be adjusted to meet monthly targets. Later sections will provide more detail on how this information is used. In short, systems must be implemented to provide operational and maintenance work data that facilitates program planning and scheduling adjustment. Work orders must be generated for all maintenance and repair events regardless of availability impact. Avoid the use of general "blanket" or standing work orders for smaller jobs because this will hinder later investigation of maintenance events in terms of availability. Blanket orders lose the detail and identification of specific problematic machine components or systems. The shift log can be used as a comparison to the work order system to make sure that work orders are being generated for each maintenance event.

Work order data is valuable to allow the adequate analysis and report generation. Repair events should be categorized by SMCS code. This simply allows for more detail of the work order that can be utilized for downstream data analysis. Likewise, more detail and data should be captured by accurate use of shift reports and SIMS tickets to identify cause of failure. Part of the planning function is to monitor the cost structure of the maintenance operation to ensure the correct cost balance.

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Including the parts and labor on the work order will enable planning to calculate the percent spent in each category.

The final output of this data and recordkeeping will be the management reports. These reports are used to measure and evaluate the maintenance groups' performance and effectiveness. They are related to the performance metrics and are used to evaluate if the maintenance effort is resulting in target machine availability, MTBS, MTTR, etc. If these top indicators are below target then more function specific metrics and reports can be used to determine root cause such as Service Accuracy, % Scheduled Work, Backlog completion rate, etc. Performance evaluation will be covered later in the document but the following are examples of management reports generated from the same maintenance and operations database discussed above. See Charts 1 – 5.

Top Tier Management Reports: Availability, MTBS, MTTR

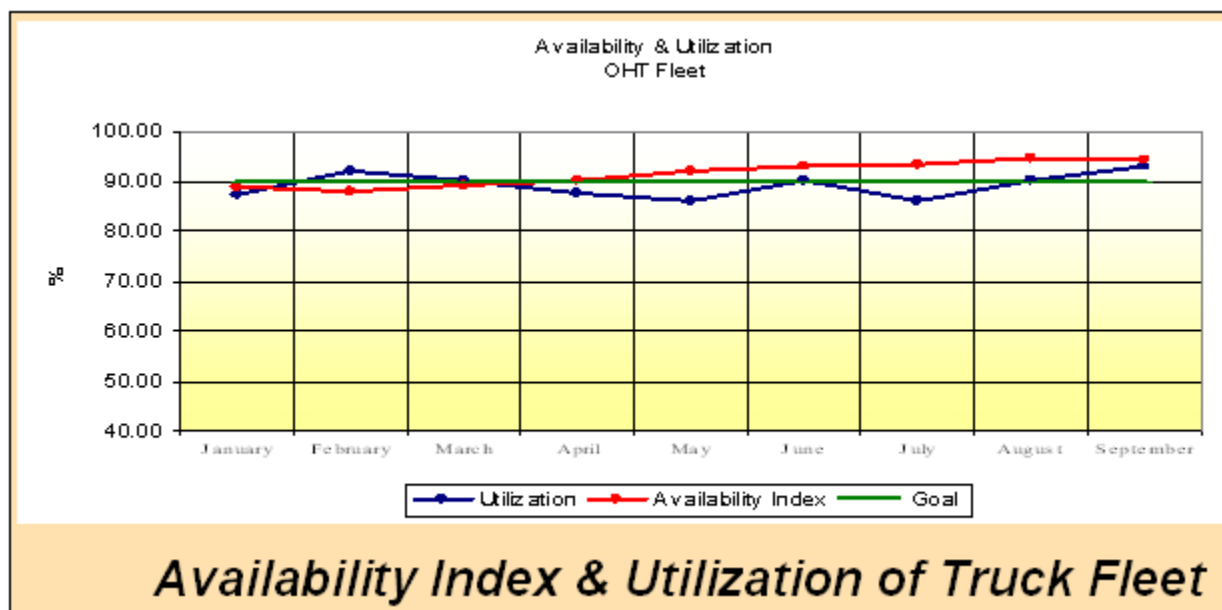


Chart 1

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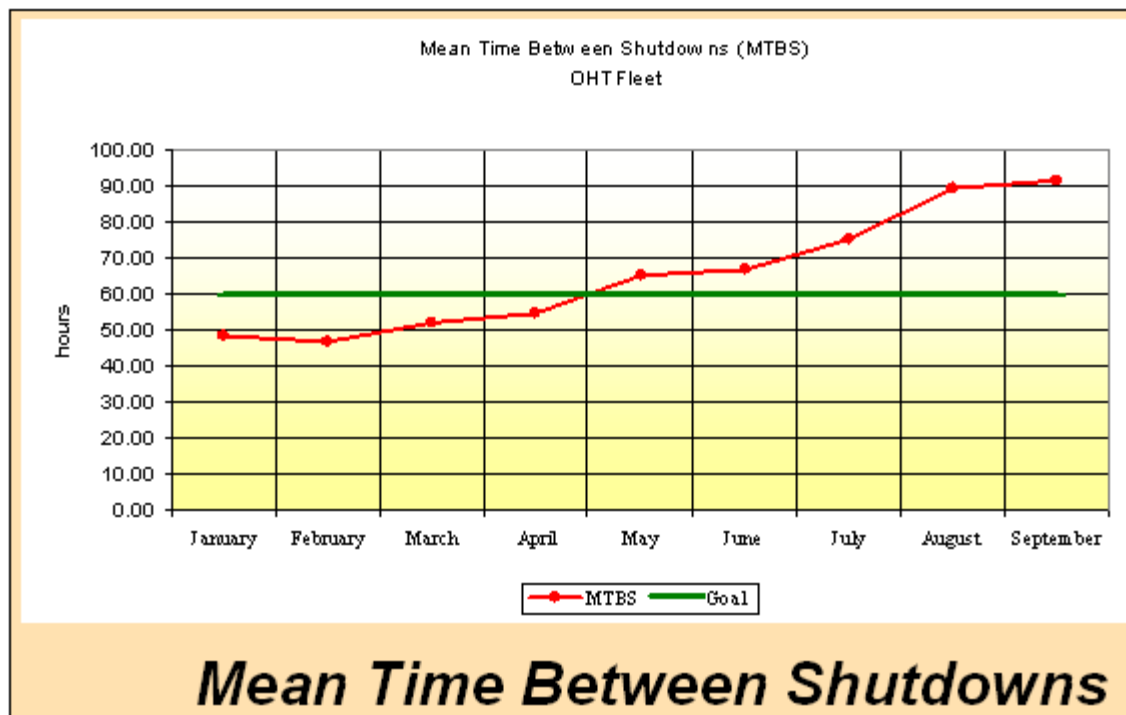


Chart 2

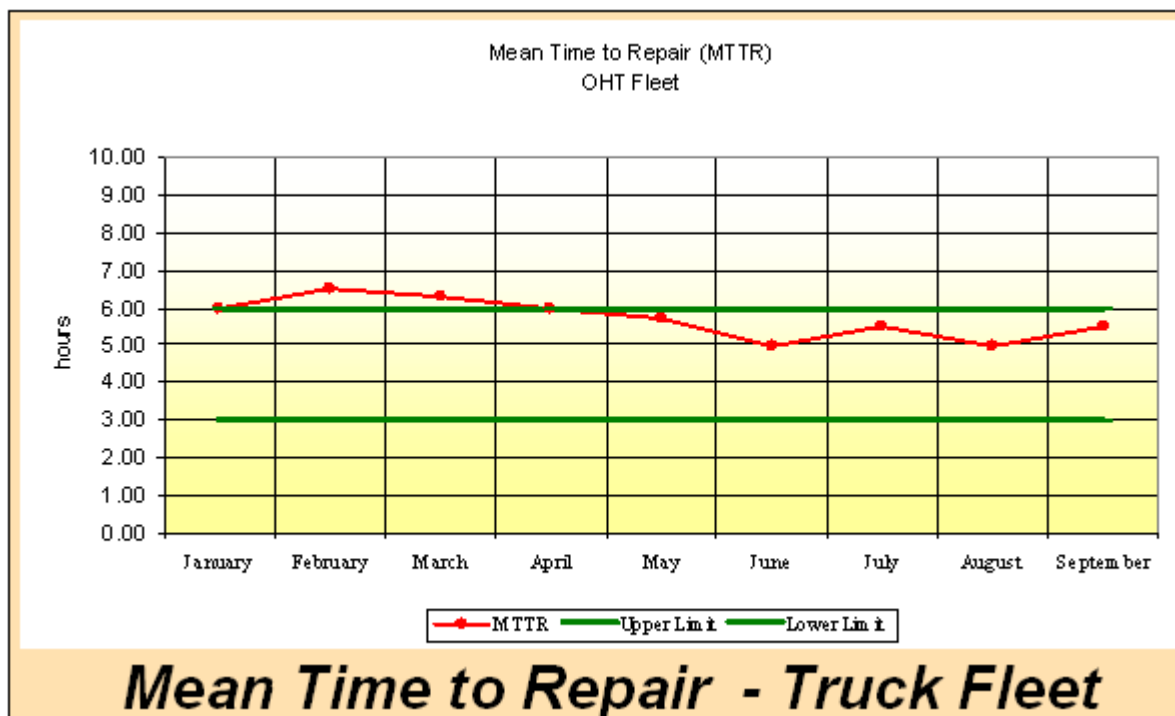


Chart 3

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Supporting Reports: Service Accuracy, % Scheduled work, Backlog generation vs. completion, etc.

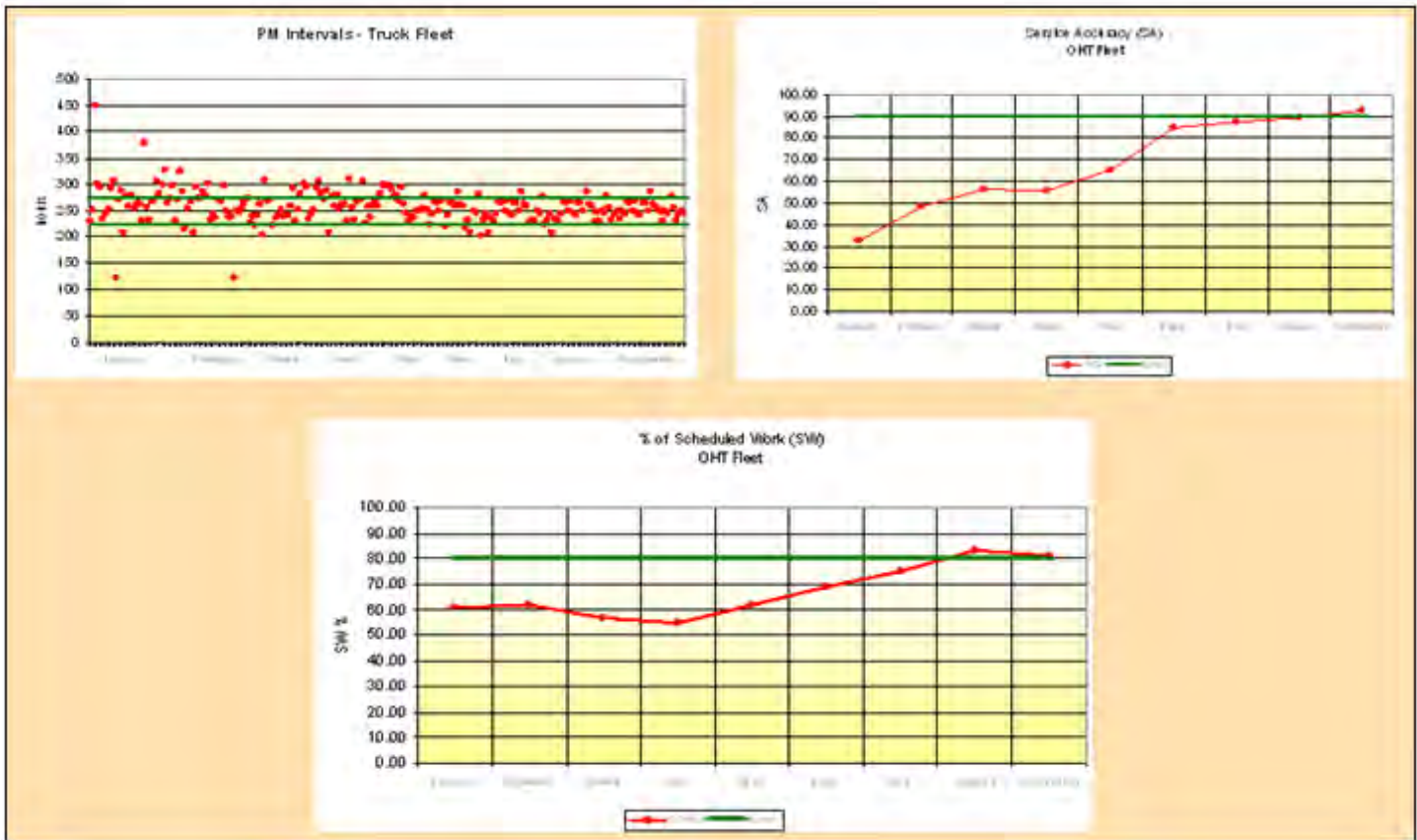


Chart 4

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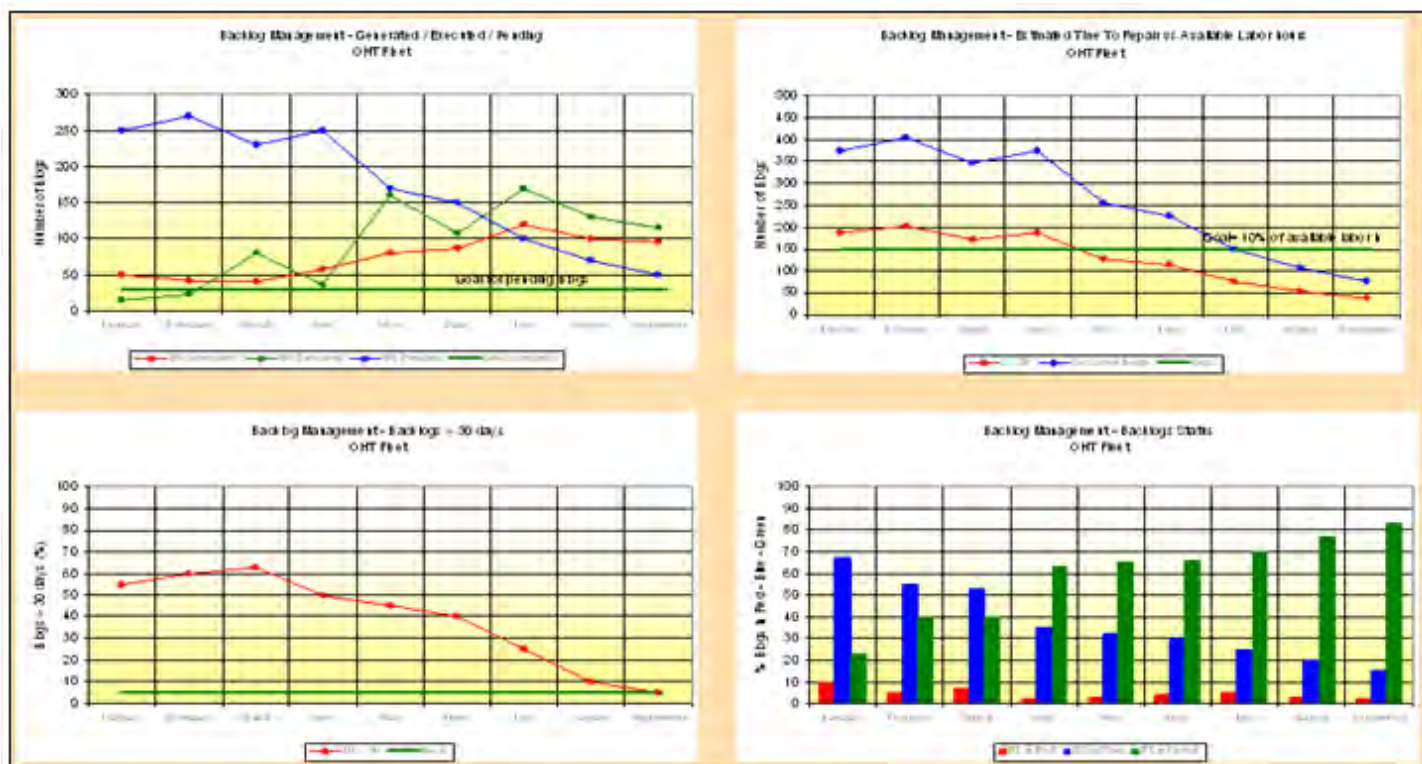


Chart 5

2.2.3 Planning

Planning must balance activities between maintenance requirements, machine availability and resources. The Planner's overall goal is to perform the required Preventive Maintenance, Backlogs, PCRs, Major and Minor repairs, and Condition Monitoring within the allowed unavailability time for each fleet. Remember that it all starts with the maintenance strategy. Now the goal is to develop this strategy into Short, Medium, and Long-term maintenance plans for the fleet and individual machines. We will use the Planned Component Replacements (PCRs), part of the Long-term plan, as an example to illustrate the planning process.

The initial PCR plan is developed for the fleet based on multiple factors. The first being factory component life projections, historical life data, and results from any previously performed MSR. As the component reaches the life target it becomes increasingly important to replace that component. This replacement urgency must be managed against fleet and machine availability target and available resources. The PCR plan will define the specific serial number machines to receive specific component replacements (i.e. torque converter) in a given week through the next 12 months and the impact on availability and resources. For example, machine 107 will be changing the engine and torque converter during week 3 of the first month.

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Table 2

Continuing on the Planner must make adjustments to potentially any plan but especially the PCR plan based on condition monitoring findings. Results from inspections, SOS, magnetic plugs, etc. may dictate that a specific component get replaced earlier than expected or is able to be delayed to accommodate a different activity. A useful tool for communication of component condition is the “Hot Sheet”.

Note: The Hot Sheet will be covered in more detail in the Condition Monitoring Best Practice (BP# 1007-2.0-1102). A basic example is shown below to demonstrate its influence with planning. See Table 3.

For example, by highlighting the machine component in red, the Hot Sheet quickly identifies the components needing immediate attention potentially sooner than the PCR plan defines. The Planner then should compare these results with his PCR plan to determine required changes.

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Equipment	Systems / Components								
	Engine	Trans	Drum	T-Connector	Throttle Cable	Overhaul	Hydraulic Oil	Leakage Oil	Other
101			Engine						
102									
103								Monitor	
104	Monitor				Engine	Monitor		Monitor	
105						Engine			
106	Engine		Engine						
107			Engine	Monitor		Monitor			
108			Engine	FL & RFL		Monitor			Monitor
109				Engine					
110								Monitor	

HEAT SHEET TERMINATOR

12 Deg. C 6 Deg. C Tot. Temperature 10 Deg. C

Table 3

Now the Planner is ready to present the Medium-term plan to the rest of the maintenance team (Project Manager, Parts Manager, Shop Manager, Scheduler, Fleet Analyst, etc.) for preparation and initial approval. Each representative will compare the PCR plan to their current commitments, resources, and capabilities to determine how the PCR plan will fit into other work and obligations. For example, the Scheduler will begin to integrate the PCR plan with his current schedule of PM's and scheduled repairs. If adjustments are needed, modifications to the PCR plan can be made at this time.

The next step is to determine the other maintenance activities that can be combined with the PCR while the machine is down. The Planner can take advantage of this window of opportunity to include additional Backlogs and Minor/Major repairs in parallel with the PCR in order to minimize the overall impact on machine downtime (availability). Once the overall PCR program is complete the Planner will submit it to the Scheduler and a Pre-PCR inspection will be scheduled to validate the specific machine PCR program. The Pre-PCR inspection process, goals, and objectives will be covered extensively in the Condition Monitoring BP# 1007-2.0-1102. It is basically a validation exercise of the maintenance activities and any special machine health improvements needing planned during the PCR.

Now that the PCR program is complete the Planner can prepare the Service Packages to aid in the PCR execution. Service Packages are a culmination of work orders, Gantt charts, standard jobs descriptions, and technical or special instructions needed for the repairs. These are documents provided to the shop foreman and technicians to guide their activities and work flow during the PCR.

One of the Planner's last but most complex and critical activities is to determine the final impact on machine downtime and maintenance resources but also expected machine performance after the PCR. Machine downtime is managed using the "Unavailability Distribution" concept where the machine's total allowed monthly downtime is portioned among the various maintenance events. In this case, the Planner wants to make sure that the PCR time planned for this month will not exceed

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the PCR allowance of 2.0%. If the 2% is exceeded then one of the other maintenance activities must be adjusted down to compensate. See Table 4.

Note: Availability Management and the Unavailability Distribution process detail will be covered in section 2.2.6.

Unavailability Distribution Goal		
Preventive Maintenance (PM)	3.5	%
PCR	2.0	%
Detection Routines	0.5	%
Scheduled Repairs	0.0	%
Unscheduled Repairs	5.0	%
Total projected unavailability	11.0	%
Max % of unavailability	10.0	%

Table 4

Similar to the unavailability impact, the Planner must determine the labor and facilities resources required for the PCR and how it may impact other planned work. For example, there are a finite number of labor hours available during a given day, week, and month. The Planner must organize the planned work with consideration of available labor to complete the work. Notice that labor estimates are part of the example PCR plan spreadsheet shown at the beginning of this section.

Looking ahead the Planner must consider how he or she will measure the success of the M&R activities, especially for a PCR, which is one of the most extensive and resource-demanding functions that the M&R team will execute. The Planner must measure how effectively and efficiently his team performed the execution of the PCR. He can achieve this evaluation through the use of metrics. One of the Planner's last responsibilities is to define, gather data, track, and analyze these M&R performance metrics. Performance Evaluation will be covered more thoroughly in Section 2.2.7 but following are a few examples.

For example, the Planner will want to measure:

Availability after PCR: Considering the amount of PCR investment, the customer should see a significant improvement in machine reliability. Setting a post-PCR availability target will indicate the effectiveness of the PCR repairs and if the correct components and systems were addressed during the PCR to yield significantly improved availability.

MTBS after PCR: Will measure the PCR effectiveness with a focus on if the pre-PCR inspection and evaluation process added value by determining the chronic system and component problems that caused past frequent shut downs.

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MTTR_{PCR}: Will measure the efficiency or preparedness of the M&R team to execute the specific PCR plan. This metric will help determine if the work plan was realistic and accurate in terms of labor and time allowance. On the other hand, it may indicate that the work plan was good but the execution did not follow the plan, i.e. work sequence, labor reallocation, parts shortages, etc.

These are just three of many KPI's used to evaluate M&R performance. The point here is that the Planner must define the metrics to be utilized to evaluate his future M&R activities, in this case a PCR.

The very last step for the Planner is a final review of the PCR plan with the M&R team; i.e. the Project Manager, Parts Manager, Shop Manager, etc. Once final approval is given by each member regarding resources, cost, PCR scope, etc., the Planner can release the job to the Scheduler to be executed.

Overall the Planner must define the Short, Medium, and Long-term maintenance plans with consideration to Condition Monitoring results and their required adjustments to the plan. The Planner must also set goals for the monthly completion rates of backlogs. He will be tracking the total number of backlogs pending per machine versus his monthly target. For example, if the backlog creation rate is higher than the execution rate then the total pending will increase. The Planner must set the target completion rate for the Scheduler in order to achieve his monthly target for pending backlogs per machine. Ultimately the Planner is responsible for defining all maintenance activities and developing plans for the Scheduler to execute while evaluating performance of the overall M&R organization and making the appropriate performance improvements.

2.2.4 Scheduling

The main function of Scheduling is the execution of the M&R plan. The Planner defines all M&R activity requirements but it is Scheduling that determines when these activities will occur and how they will be combined to maximize resources, minimize downtime, etc. The scheduling is accomplished through the creation, communication, and facilitation of a rolling 7-Day Short Term Program. Initially it may seem simple to populate a weekly calendar with M&R activities but the complexity occurs when these activities must be balanced with machine-specific and fleet availability, available labor, and resources.

The following section discusses the five scheduling stages required to achieve the final balanced 7-Day program.

Stage 1: Schedule PM events and Backlogs in parallel.

The best starting point for populating the 7-Day program is the Preventive Maintenance (PM) events because they are well-defined, known interval, high frequency activities. Inputs will be the defined PM intervals from Planning, estimated machine hour accumulation and actual hour meter readings. With these inputs, the Scheduler knows current machine hours and can accurately estimate when the machine will need to be serviced to meet its PM interval target.

Due to high PM frequency and consistency, these operations become a prime window of opportunity to perform backlog repairs in parallel. The planning department should be setting backlog execution targets based on scheduled PM downtime, portion of PM downtime reserved for backlog completion,

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and estimated time per backlog. For example, if PM downtime is 10 hours with 40% reserved for backlogs then 4 hours are available for backlog repairs. With a historical backlog repair time of 1 hour the Scheduler should target 4 backlogs per PM. By reviewing the current backlogs “ready to go”, priority, and estimated time to repair the appropriate backlogs can be chosen and placed in the 7-Day program.

Note: Backlogs scheduled are always subject to change based on the backlog validation results from the Pre-PM inspection.

At this point the Scheduler has populated the 7-Day program with PM's and machine specific backlogs to be performed during the PM downtime. He is now ready to move to Stage 2 which adds the other M&R events in an attempt to occupy any remaining shop bay time, labor etc.

Stage 2: Incorporate PCR's and other Scheduled Major & Minor repairs.

Again, this input comes from Planning in the form of impending PCR's. As discussed earlier in the Planning section, the Planner will submit a machine-specific PCR plan based on accumulated component hours and recent Condition Monitoring feedback (Hotsheet method). It is the Scheduler's job to add this PCR to the 7-Day program in coordination with the already scheduled PM's and backlogs. A PCR typically will occupy bay space and resources for multiple days therefore it will consume a relatively large portion of the 7-Day program.

The last remaining M&R event for consideration is a Scheduled Major or Minor repair. These simply are existing backlogs that require either a separate stop (outside a PM stoppage) or a PM downtime extension. There are two possible reasons for this event: first is priority. For multiple reasons, the risk for failure might be too high to operate the machine until the next PM opportunity to perform the repair. Therefore, a new scheduling window must be opened to down the machine for repair. A second scenario could arise from a situation where the backlog creation rate is surpassing the backlog execution rate and new scheduling windows must be opened to catch up on pending backlogs. Ideally, Planning will establish a monthly backlog execution goal that gains control of the pending backlog workload over a targeted time period. In either case, Planning will define this requirement and the Scheduler must eventually accommodate this additional Scheduled Repair within the 7-Day plan.

At the end of Stage 2, the 7-Day program will resemble the following example with PM's, PCR's, and backlogs populating the schedule. See Table 5.

Time	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
05:00	110 - PM2	108 - PM2	103 - PM1	106 - PM1	101 - PM2		
07:00							
09:00			109 - PM1	108 PCR	108 PCR	108 PCR	108 PCR
11:00		107 - PM3					
13:00	105 SR						
15:00							
17:00							
19:00							
21:00							
23:00							

Table 5

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Stage 3: Balance Downtime & labor required with Availability goal and labor available.

Now that the Scheduler has populated the 7-Day program he must make sure the schedule is not in violation of fleet availability targets and available labor resources. For availability: (reference to Table 6 below), the total machine downtime hours are calculated for each day. This is a tally of known scheduled downtime and estimated unscheduled downtime based on historical work order data. With daily downtime hours established, the Scheduler can estimate a daily fleet availability. In the example below, Friday's estimated availability is 88%. If the availability target is 85%, then the schedule is above target and has room for adjustment if needed (i.e. shifting work from a different day to Friday).

Time	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
05:00	110 - PM2	106 - PM2	103 - PM1	106 - PM1	101 - PM2		
07:00							
09:00			109 - PM1	108 PCR	108 PCR	108 PCR	108 PCR
11:00		107 - PM3					
13:00	105 SR						
15:00							
17:00							
19:00							
21:00							
23:00							
Downtime - Availability Projection							
PM h	8	16	16	8	8	0	0
Backlogs	2	4	8	0	0	0	0
# Backlogs							
Scheduled h	8			20	24	24	24
Unscheduled h	12	12	12	12	12	12	12
Total h	30	32	36	40	44	36	36
Est. Availability	88%	87%	85%	83%	82%	85%	85%
Labor Hours Requirement - Definition							
MH - PMs	40	80	80	40	40	0	0
MH - Backlogs	12	24	32	8	8	0	0
Scheduled h	24	0	0	60	72	72	72
Unscheduled h	48	48	48	48	48	48	48
MH - Required	124	152	160	156	168	120	120
MH - Available	150	150	150	150	150	150	150

Table 6

Likewise, the same exercise is done for labor hours. In this example, 150 hours of labor are available each day. It can be seen that the schedule for some days exceeds the 150 hours available and therefore jobs must be shifted to days with lower labor estimates.

While focusing on the weekly balance, the Scheduler must also track monthly availability and labor impact. The two tables below illustrate the methodology of translating fleet availability to available M&R hours. By multiplying the fleet production plan hours by the availability goal (90% in this case) the Scheduler can estimate his available M&R hours for the month, in this case 720 hours (see Tables 7 & 8). As the M&R work continues through the month the schedule must deduct the used hours from the total available for the month. The main reason for this tracking is to make sure he will not consume too many hours early in the month and leave insufficient hours for the end of the month activities. If nothing else, the schedule must always continue to accommodate PM's. If he does not reserve enough hours for the end, then the M&R hour total will exceed 720 and availability will not make the 90% target.

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Consult Fleet Management Goals	
Fleet Management Goals (Month)	
Number of Machines	10
Physical Availability Goal	90 %
Utilization of Asset	85 %
Hours available for Maintenance & Repair	720 h

Table 7

Compare with available (remaining) hours for M&R	
Maintenance hours Tracking (Month)	
Total hours available for M&R	720 h
Hours used up to date	455 h
Remaining hours for M&R	265 h
Availability Management	

Table 8

Graphical representation is also a helpful tool to balance workload between weeks of the month. In the example below it becomes apparent that the current schedules for week 1 and 2 are heavy on work. The lower workload (higher projected fleet availability) of week 3 makes it a good option to accept activities from week 1 and 2. Note that the Scheduler must consider more than just availability and labor allowances. In the bottom two graphs, Service Accuracy shows PM execution compliance. Significant changes to the PM schedule to meet availability or labor targets will adversely affect Service Accuracy compliance. Everything is a balance. See Chart 6.

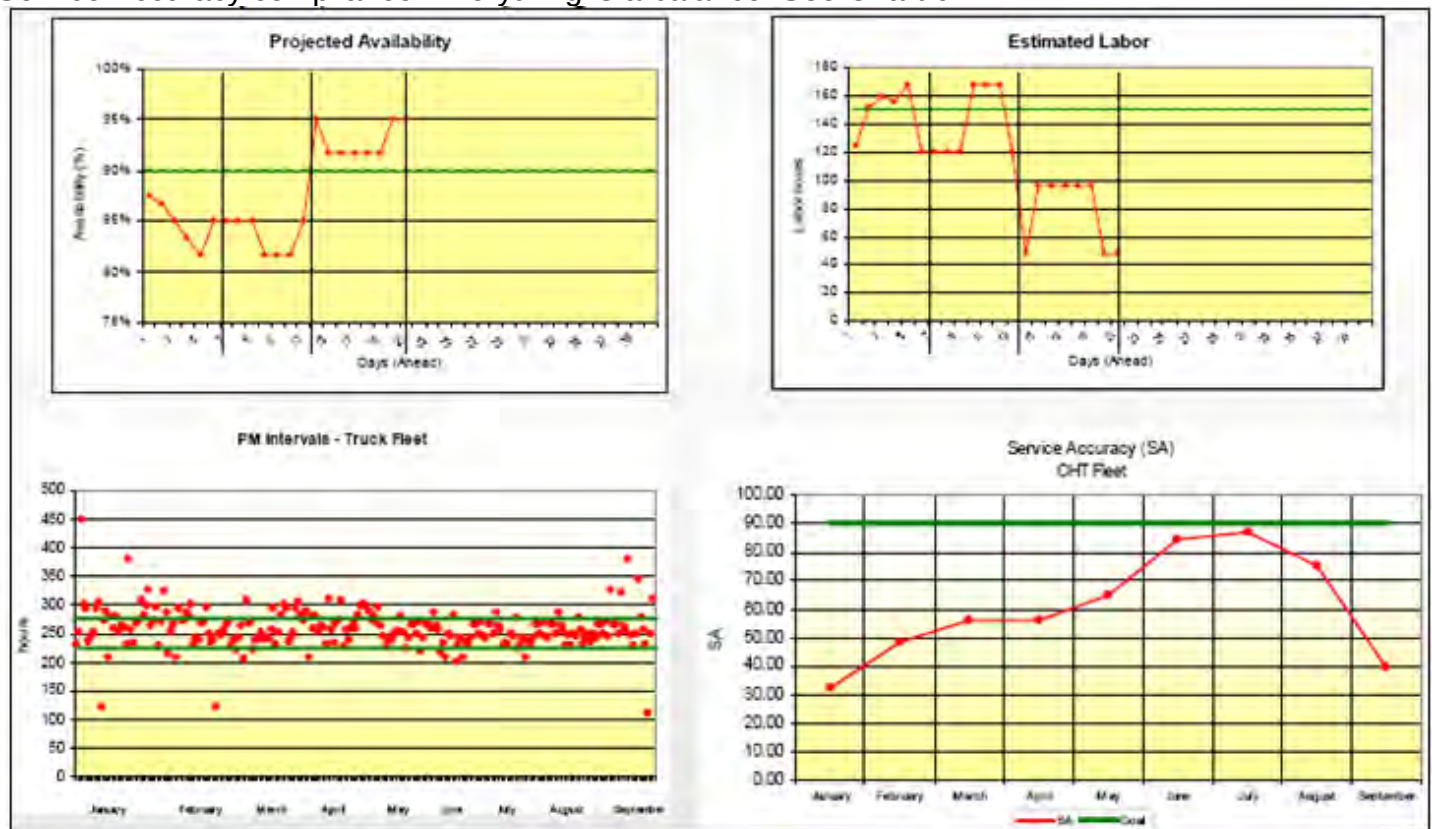


Chart 6

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The Scheduler will be working closely with the Planner to authorize program adjustments due to availability or labor constraints. When a balanced 7-Day program is finalized, the Scheduler will initiate orders for logistics; external & internal labor, PM kits, backlog parts, special tooling, etc.

Stage 4: Share and communicate the schedule to the organization.

To ensure agreement and successful execution, the final schedule must be reviewed with the appropriate staff and management personnel. The management meeting will be at a higher strategic level with the Production Manager, Project Manager, Customer Maintenance Manager, Planner, etc. A second more tactical meeting should occur with the Fleet Analyst, Parts Supervisor, Shop Manager, Planner, etc. In both meetings the intent is to find potential conflicts, make modifications if necessary and finalize the upcoming schedule. Once consensus has been reached, the program is published for execution to production, the shop, and the field and support departments.

Stage 5: Control execution/completion of the program.

The most dynamic aspect of the Scheduler's responsibilities is tracking, adjusting and executing the current 7-Day program. It's a very fast paced activity because of the multitude of real-time data to track from multiple mine organizations. Usually the first step is to review of the work progression for the previous day via work orders or the shift log report. The Scheduler must determine if the repair and PM shop is ahead or behind schedule and how the schedule must be adjusted. Changes to the current day's or week's schedule must immediately be communicated to production dispatch. If the shop is ahead of schedule, the Scheduler must determine what jobs can be added and request from dispatch that the appropriate machines be pulled from production. Likewise, if shop work is behind, the Scheduler must request production dispatch to delay the delivery of the scheduled machines. Continuous job completion, follow up, and making the appropriate adjustments is essential to maintain the most effective and efficient use of shop resources and machine production uptime. Towards the end of shift, the Scheduler must update the 7-Day program with the daily changes and review/finalize these changes with the management or tactical meeting participants discussed above. Therefore everyone is clear on what the schedule originally was for the day, what was executed, the required adjustments and implications, and the revised schedule for the rest of the week.

2.2.5 Technical Support

As shown in the "General Process" map above, the Technical Support function or group is one of the seven critical areas required for a successful Planning and Scheduling department. The Technical group is responsible for analyzing incoming information pertaining to the local equipment health and current product problems. They provide overall technical support of the planning and the subsequent shop execution of the maintenance and repair requirements. They communicate technical requirements to planning and act as a technical reference for shop operations.

The Technical groups' main responsibility to Planning is to provide the latest factory and local support/rework program requirements, i.e. PIP's, PSP's, local dealer solutions, etc. Providing the parts, labor, and time details allow the Planner to plan the job within the existing ST, MT and LT (short, medium, long-term) programs. The Technical group can enter a Backlog request to Planning if the job is to be completed in the medium to short term, or coordinate it with Planning as a Scheduled

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Repair in the long term plan. Lastly, the technical group should also track the completion of these programs on a serial number basis to assure fleet compliance.

Another function of the group is to provide technical guidance to the M&R operations side. The main shop, PM shop, and field mechanics require support related to scheduled and emergency repairs. The Technical group shall provide effective support of requests regarding parts, repair instructions, special tooling, etc. They shall also be the liaison to the dealer and factory technical personnel.

Over time the Technical group will accumulate much machine health and product problem information through networking with factory and fellow dealer personnel, and also developing problem solutions internally. As a means to facilitate quick information retrieval, the Technical group must develop and maintain a solutions database to retain resolution definitions and instructions.

Technical communication channels need to be identified and used to facilitate fast and accurate data transfer to Planning and Repair Management. Part of this efficiency could come from having a computerized data and document system for the Technical group to reference. Requests to the group can accumulate and sometimes be difficult to manually manage. The use of a request tracking system will free up the technical personnel to concentrate more time on providing data and reduce the chance of requests being lost.

2.2.6 Availability Management

The overall goal of Planning and Scheduling is to accomplish the required M&R activities during the two categories of machine stoppages: planned periods of fleet unavailability and any operational windows of opportunity. Fleet unavailability is determined by the CSA or MARC availability commitment between the customer and dealer. Operational windows of opportunity are based upon the mine production department goals for machine utilization. In the following example (Table 9), the availability goal is 89%, which leaves 11% of the machine time available for M&R activities. Similarly, the Utilization of Asset defines the operational window of opportunity. This is the time that the customer has the machine available for production, but is not actively using the machine. In this example, the Utilization of Asset is 85%, which yields an additional 4% of machine time potentially available for M&R activities.

Note: the operational window of opportunity represents the time that the machine is available for production and is not being utilized but is still controlled by the customer. Therefore the dealer's window of opportunity is strictly dependent on the customer's willingness to allow the dealer access to the machine during this unutilized time. In other words, it's not guaranteed M&R time.

Fleet Management Goals (Month)	
Number of Machines	10
Physical Availability Goal	89 %
Utilization of Asset	85 %

Table 9

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At this point, Planning and Scheduling has defined large portions of M&R time. In order to best manage and utilize the available time for M&R activities, the Unavailability goal and operational window of opportunity must be broken down into available hours. Continuing with the example, the 11% unavailability allowance translates into 792 available M&R hours per month (See Table 10). As the month progresses, Planning and Scheduling will tally the hours consumed via work orders and shift log report history. This allows P&S to track the remaining M&R hours available through the end of the month.

Maintenance & Repair hours Tracking	
Date: Monday xx,yy,zz	
Total hours available for M&R	792 h
Hours used up to date	455 h
Remaining hours for M&R	337 h

Table 10

This hour breakdown is a good start but is still too general because it represents all hours for all M&R activities combined. In order to make adjustments during the month, the analysis must break down and track both the hours consumed and remaining *per each M&R activity* (i.e. PM, PCR, CM, scheduled and unscheduled events).

The first step is to breakdown the 11% unavailability allowance. We call this the Unavailability Distribution Goal. It should be defined within the P&S Strategy and Plan but will change as fleet and application characteristics change. Below is an example distribution of the 11%. A portion of the 11% is allocated to each of the M&R activities (See Table 11). This will enable the Planner and Scheduler to more accurately analyze the time spent per activity.

Next, the Unavailability Distribution Goal is combined with the Unavailability Hour Goal to yield hours consumed and remaining for each of the M&R activity categories. The last chart shows this hour distribution. See Table 11.

We have now developed a useful Availability Management tool. By tracking consumed and remaining M&R activities hours, Planning and Scheduling can manage and adjust the monthly and weekly program to meet the distribution goals and assure appropriate time is spent for each M&R activity. In some cases, the consumed hours will approach or exceed the total hour goal for that activity. For example, detection routines (CM) have used 37 hours versus the 36 hours targeted by the distribution (see Table 11). In this case, P&S would consider the operational windows of opportunity to execute required detection routines for the rest of the month. The operational hours will complement the unavailability hours. As described in Stage 3 of the 2.2.4 Scheduling section, this is simply one more tool to facilitate P&S's balance between machine downtime and available labor.

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Unavailability Distribution Goal					
Preventive Maintenance (PM)	3.5	%			
PCR	2.0	%			
Detection Routines	0.5	%			
Scheduled Repairs	0.0	%			
Unscheduled Repairs	5.0	%			
Total projected unavailability	11.0	%			
Max % of unavailability	11.0	%			

M&R hours Tracking - Unavailability Distribution Goal					
Date: Monday xx.yy.zz					
			Avail h	Used h	Used %
Preventive Maintenance (PM)	3.5	%	252.0	114.0	45.2
PCR	2.0	%	144.0	0.0	0.0
Detection Routines	0.5	%	36.0	37.0	102.8
Scheduled Repairs	0.0	%	0.0	0.0	0.0
Unscheduled Repairs	5.0	%	360.0	304.0	84.4
Total projected unavailability	11.0	%	792.0	455.0	57.4
Max % of unavailability	11.0	%	792.0		

Table 11

2.2.7 Performance Evaluation

Long term effectiveness and improvement of the Planning and Scheduling department must begin with a formalized performance measurement system. Based on the performance results, contributing factors can be determined and improvement plans can be developed and implemented. Effectiveness of the improvement plans is evaluated by the same performance metrics and the cycle continues. The following is a general methodology to measure and analyze the performance of the Planning and Scheduling group, along with the M&R organization's compliance to the program. The end goal is to identify positive and negative contributors to the top-tier metrics and develop improvement plans to address the underperforming areas of the M&R organization.

The M&R organization's ultimate goal is to meet the fleet performance top-tier metric targets. Therefore, start with gathering data and calculating the top tier metrics to measure the overall fleet results. The first "performance check" is to compare results to previously established targets. This will expose the underperforming top-tier metrics. These underperformers will require further investigation into the contributing lower-tier metrics to find root cause. The following example lists the recommended top tier metrics. See Table 12.

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Fleet Goals	
Availability	_____ %
MTBS	_____ h
MTTR	_____ h
Service Accuracy	_____ %
PM Execution Time	_____ h
MTBS after PM	_____ h
Backlogs / Machine	_____ -
BL to execute / PM	_____ -
BL to exec. / month	_____ -

Table 12

Full definition of all metrics mentioned can be found in the “Metrics (KPI’s) to Access Process Performance” document attached, and found at CatMiner.com under Maintenance & Service. **The KPI document includes each metric’s description, formula, contributing metrics, and interpretation guidelines. Utilize this document as a guide to identify the under-performing contributing metrics which will eventually identify the area of the M&R organization that either lacks execution compliance or requires process change.**

Perhaps the one metric that will give the best indication of the M&R organization’s “level of control” of fleet condition and performance is % Scheduled Downtime. This a simple ratio of scheduled downtime hours and total overall downtime hours. This metric reflects the Condition Monitoring, Preventive Maintenance, and Backlog Management’s ability to detect a defect and effectively plan, schedule, and execute a repair before the defect results in an unscheduled repair. It measures the effectiveness of the M&R organization’s **preventive** maintenance approach of repairing before failure. The benchmark target is 80% scheduled downtime. The nature of mobile equipment allows for a certain amount of unscheduled downtime (thus the 20% benchmark). Unscheduled downtime will be minimized as the M&R routines and execution improve.

One area to investigate for potential problems that could be affecting top tier metric performance is the Unavailability Distribution mentioned in the previous section. Remember that the Planning group’s goal is to develop an M&R plan that achieves the top tier fleet performance targets. Shown below is a comparison between the Distribution Targets and Monthly Results (See Table 13). Deviations from the targets raise many questions. Is the plan realistic to properly maintain the fleet? Is the schedule following the plan? Are the executors complying with the plan and/or schedule? Did we plan for efficiency? Did we schedule all needed resources? Again, metrics will lead the analyst to potential problems areas. Further investigation is always needed to find root cause.

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Unavailability Distribution Results			Targeted			
Preventive Maintenance	3.5	%	3.0	%	Diff: 0.5 %	
PCR's	3.4	%	1.5	%	Diff: 1.9 %	
Major & Minor Repairs	0.7	%	1.0	%	Diff: 0.3 %	
Detection Routines	0.3	%	0.0	%	Diff: 0.3 %	
Tires	3.2	%	1.5	%	Diff: 1.7 %	
Unscheduled Events	6.2	%	5.0	%	Diff: 1.2 %	
Total	17.3	%	12.0	%		

Table 13

Another good starting point to evaluate the M&R organization's basic ability to plan, schedule and execute activities is Service Accuracy. The plot below shows the actual PM execution intervals relative to the 250 hr target (See Chart 7). Considering that Preventive Maintenance is the most basic and frequent routine, it serves as a good initial review of the organization's capability to schedule and execute work on time. Assuming the PM's are defined and scheduled accurately, a poor Service Accuracy metric will point to the shop and PM crew's preparedness and compliance to the schedule as the root cause.

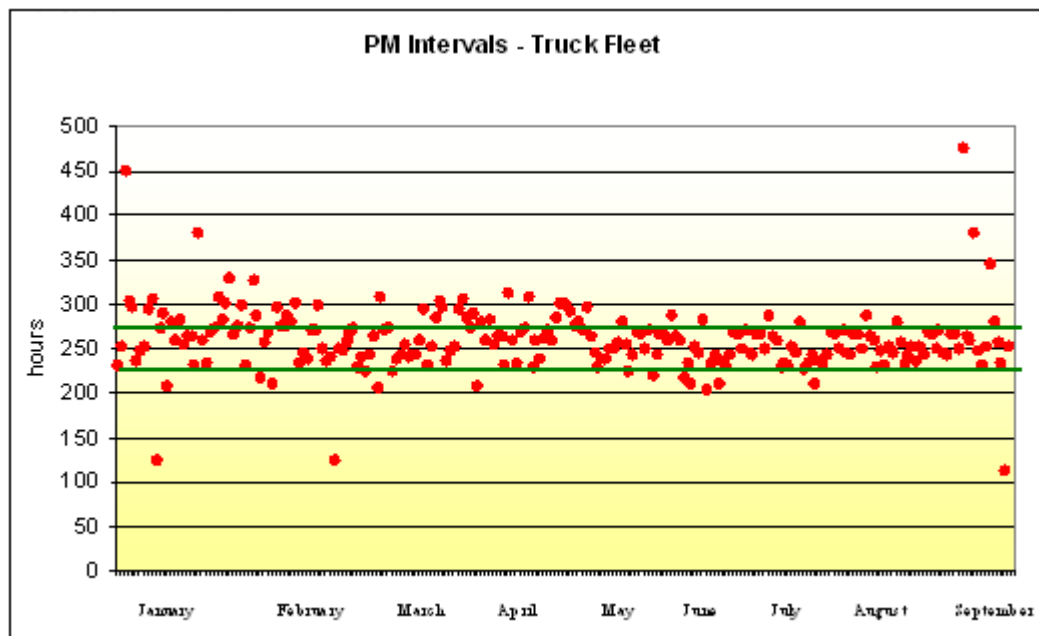


Chart 7

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After the top tier metrics are defined and targets established, the last few examples were given as good starting points for indicators of performance. Performance Evaluation is an extensive topic and can be increased in detail as required by the top metric results and indication to improve. As mentioned earlier, the objective is to first evaluate the state of the top tier metrics and expand the analysis deeper into the supporting metrics until the root cause or underperforming areas are identified. For example, each M&R process can be evaluated through multiple levels of metrics. See Table 14.

Preventive Maintenance PM Performed / PM Scheduled _____ Total PM downtime scheduled _____ Actual PM downtime used _____ Total PM labor hours scheduled _____ Actual PM labor hours used _____ Projected MTBS after PM _____ Actual MTBS after PM _____ Backlog Management Backlog executed / Backlog scheduled _____ Backlogs scheduled for PM _____ Backlogs executed in PM _____ Backlogs executed / PM _____ Backlogs scheduled for the Shop _____ Backlogs executed in the Shop _____ Backlogs executed in Field _____ Total labor hours scheduled for Backlogs _____ Total labor hours used for Backlogs _____ Total Backlogs executed up to date _____ Major / Minor Scheduled Major components executed / scheduled _____ Major overhaul executed / scheduled _____ Partial repairs executed / scheduled _____ Labor hours scheduled for scheduled repairs _____ h Labor hours used for scheduled repairs _____ h Overall PCR completion rate _____ %		Major / Minor Scheduled Repairs Major components executed / scheduled _____ % Major overhaul executed / scheduled _____ % Partial repairs executed / scheduled _____ % Labor hours scheduled for sched. repairs _____ h Labor hours used for scheduled repairs _____ h Overall PCR completion rate _____ % Unscheduled Repairs Projected unscheduled % of downtime _____ % Actual unscheduled % of downtime _____ % Projected unscheduled stop hours _____ h Actual hours of unscheduled stops _____ h Planned labor hours for unsch. stops _____ h Actual labor hours used in unsch. stops _____ h Program Accomplishment Total actual events / scheduled _____ % Total actual downtime / scheduled _____ % Total actual labor hours / scheduled _____ % Major Corrective Actions	
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Table 14

The final goal of Performance Evaluation is to identify the problem areas, develop improvement action plans, implement, and measure improvement. Applying performance metrics is the first step in evaluating the M&R processes and their execution. Subsequent improvement falls under the much larger subject of Continuous Improvement.

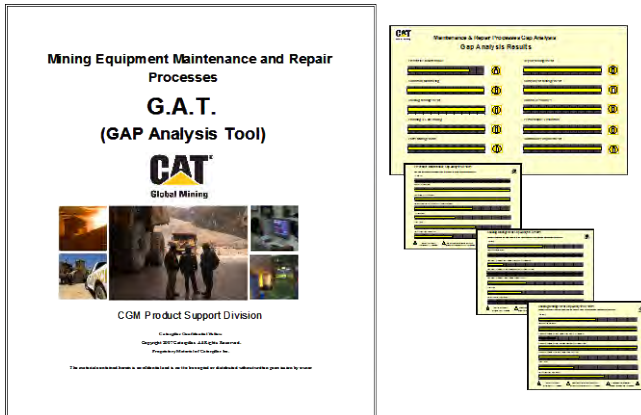
3.0 Implementation Steps

The implementation steps to follow will depend on the current situation of your mine site operation. As a first step, we recommend a thorough evaluation of your maintenance operation to determine the "As Is" situation. Compare this with the recommended model. Determine areas of opportunity and establish an action plan to correct or improve those areas.

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1. - Evaluate your on-site maintenance operation

- Apply GAT (Gap Analysis Tool) - available through Global Mining Product Support division



The GAT is a survey that evaluates the 10 M&R Processes recommended by Caterpillar Global Mining. Utilize this tool to concentrate on Planning and Scheduling.

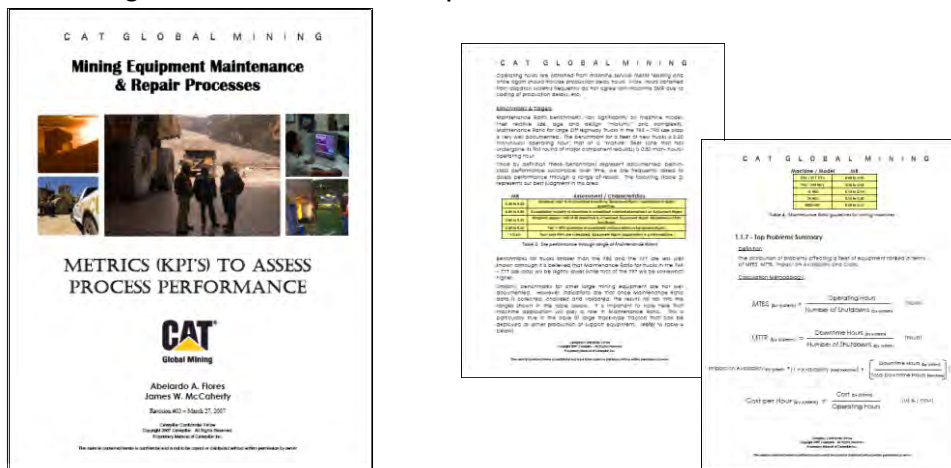
2. - Establish the “As Is” situation

- Determine the Process Flow Diagrams of all your processes implemented on site with focus on Planning and Scheduling.

Based on the observations guided by the GAT tool and the on-site investigation, draw and document the current processes on site.

- Establish the Baseline.

Measure the current performance results and document them for future reference. Apply the set of metrics recommended in CGM document dedicated to “KPI's for evaluating Maintenance and Repair Processes”



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This document contains the KPI's included in the GAT, calculation methodology and interpretation guidelines.

3. - Determine the areas of opportunities

Determine the areas of opportunities by comparing the “As Is” situation with the recommended processes detailed in this P&S Best Practice.

4. - Define the Action Plan

- Document your findings. The use of a 3W form (Who will do What by When) is recommended to record your findings and opportunities.
- Devise and document the solutions for improving or correcting the weaknesses. Use the related BP's to support your search for ideas/solutions. Use a separate column in your 3W form to record the solution/actions to take.
- Analyze and determine resources needed, time to implement, and expected benefits.
- Prioritize the implementation of the different actions based on the Return On Investment (ROI) results.
- Assign responsibilities and expected completion dates for each of the activities.

5. - Get the “Buy-In” of the on site organization

It is very important to get the buy-in of the on-site organization. They are the ultimate “process owners” of the Maintenance & Repair strategy including Planning & Scheduling. Present the action plan, the implementation strategy and the expected benefits. Involve them in the implementation of the solutions and plan an in-depth training of all the involved personnel.

6. - Implement

Execute the implementation plan.

7. - Follow up

Follow up closely with the implementation of your action plan.

- Set up “check points”
- Review compliance of committed dates for tasks completion
- Receive & analyze work progress from the specific responsible
- Reinforce the positive results
- Identify and eliminate threats and obstacles
- Support the ownership of the on-site organization
- Support the positive change of personnel, correct the negative attitudes or oppositions
- Make adjustments as necessary

8. - Measure the Results – Improve /Adjust if needed

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It is absolutely necessary to measure the results of the improvements implemented at the level of each individual processes and at the final outcome, fleet performance.

Measure and trend the behavior of the KPI's selected for establishing the "baseline" when the "As Is" situation was established. Compare and adjust as necessary.

4.0 Benefits

1. Higher reliability and machine availability
2. More efficient use of shop facilities and labor resources
3. Lower cost per hour and cost per ton
4. Improved MTTR

In general, benefits from adopting the M&R Process Best Practices including Planning and Scheduling is represented in the following slide. See Figure 5.

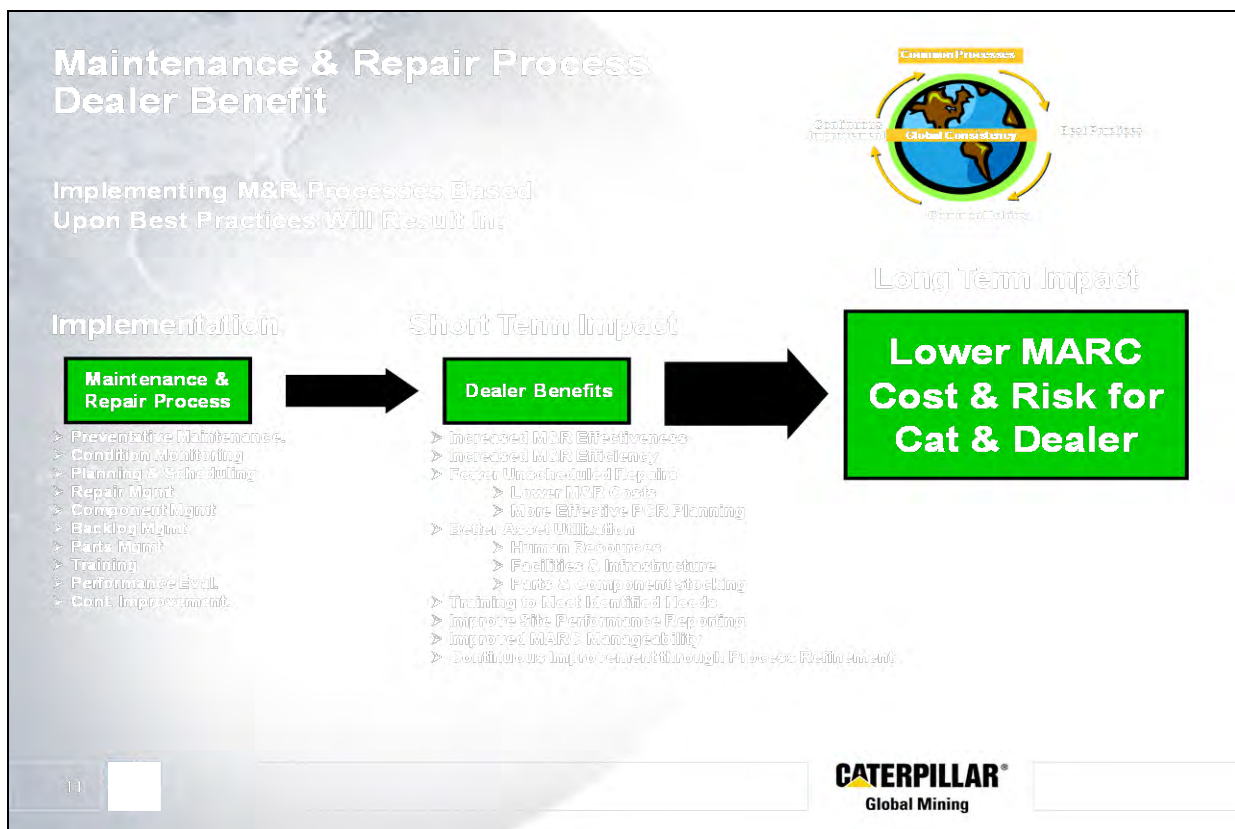


Fig. 5

5.0 Resource Requirements

• Human Resources:

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Human resource requirements associated with the execution of the Planning & Scheduling plan include:

- ✓ Fleet Analyst (Reliability Engineer) ... provides leadership and direction for all on-site Condition Monitoring activities.
 - Defines Condition Monitoring routines that correlate to equipment health, operation & application.
 - Reviews / revises Condition Monitoring procedures & checklists to ensure they focus on significant issues.
 - Ensures the accuracy & completeness of data acquired through Condition Monitoring.
 - Evaluates / interprets all Condition Monitoring data.
 - Advises Planning of issues related to equipment health, maintenance, application & operation.
- ✓ Planner ... develops plans including resource & time estimates for the M&R routines.
 - Ensures that routines, plans and activities are aligned with the strategy.
 - Defines, organizes & procures facilities, manpower, parts, tooling, forms, checklists, etc.
 - Establishes M&R activity priorities for execution.
 - Monitors maintenance and repair performance metrics for compliance with goals; devises & communicates plans to address shortfalls to the Project Manager.
- ✓ Scheduler ... translates the M&R plan into an actionable schedule.
 - Identifies “Windows of Opportunity” for execution of M&R activities.
 - Schedules the weekly program with resource availability.
 - Publishes / distributes the schedule & obtains agreement from all appropriate areas.
 - Adjusts the weekly program to comply with variations in equipment utilization & resource availability.
- ✓ Parts Analyst ... supports the M& R program by providing the right parts at the right time in the right quantity.
 - Maintains parts & components on site inventory based on planning recommendations
 - Manages the on site parts inventory
 - Implements, maintains and supplies parts kits
 - Identifies and corrects deviations
- ✓ Technical Communicator ... be the single point of contact for all technical communications.
 - Keeps fluent channels of technical communications (internal – external)
 - Provide updated technical information of on site equipment

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- Searches, acquires, and finds available solutions for technical problems
- Provides technical support to planning and service operations
- ✓ Data Entry Clerk ... maintains the Record Keeping System
 - Checks and audits all incoming information (WO, Shift Reports, etc.)
 - Enters the incoming information into the Information Management System (Computerized or Paper System) in a timely manner
- ✓ Project Manager ... is the primary on-site individual responsible for the implementation and execution of the Planning & Scheduling strategy.
 - Has total familiarity with customer expectations, contractual commitments.
 - Authorizes resources to support successful implementation of the Planning & Scheduling strategy.
 - Audits fleet and P&S process metrics; takes action to eliminate any deficiencies.
 - Authorizes revisions to P&S strategy and procedures.
- Logistical Resources:
Logistical resource requirements associated with the execution of the Planning & Scheduling plan include:
 - ✓ Work Space
 - Suitable office space to accommodate all P&S personnel and Functions. Must be organized in working areas, have enough space, promote efficiency. Location is very important, must facilitate the follow up of all work in process.
 - ✓ Meeting Room
 - Suitable meeting room where all plans are displayed and analyzed. Should be equipped with Projector, Clipboards, Boards, Bulletin Boards, etc.
 - ✓ Office Equipment
 - P&S office(s) must be equipped appropriately, avoid any delays / inefficiencies due to lack of office equipment. Computer terminals, Printers, Plotters, Fax Machines, Desks, Bulletin Boards, Flat Computer Screens, etc., are some of the equipment to consider.
 - ✓ Transportation
 - Maintenance & Repair events take place in different locations. P&S must have transportation available to check, follow up and inspect these activities. Regular visits to the Pit are also necessary and highly recommended.

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✓ Communications and Information Management System

- Efficient communication systems are a must. P&S coordination is very critical. Suggested equipment include the following systems:

- Telephone (Internal & External)
- Radio Com.
- Internal Computer Network
- External Computer Network (Connected to Central Store)
- Dispatch terminal

6.0 Supporting Attachments / References

Maintenance Strategy Reference (MSR)
 Site Assessment Tool (SAT)
 Gap Analysis Tool (GAT)
 “Metrics (KPI’s) to Assess Process Performance”

Reference material and contact names to the above can be found at CatMiner.com

7.0 Related Best Practices

1. M&R Processes ... Our Model	BP 1007-2.0-1100
2. Preventive Maintenance	BP 1007-2.0-1101
3. Condition Monitoring	BP 1007-2.0-1102
4. Backlog Management	BP 1007-2.0-1103
5. Parts Management	BP 1007-2.0-1105
6. Component Management	BP 1007-2.0-1106
7. Repair Management	BP 1007-2.0-1107
8. Human Resources / Training	BP 1007-2.0-1108
9. Performance Evaluation	BP 1007-2.0-1109
10. Continuous Improvement	BP 1007-2.0-1110

8.0 Acknowledgements

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For further details and information regarding this Best Practice, please contact the author.

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